

Chapter 5: Fundamentals of Electricity & Ohm's Law

Physics & Electrical Engineering Fundamentals

1. BASIC ELECTRICAL QUANTITIES

Electricity is the flow of electrical charge through a conducting material. To understand how electrical circuits function, three fundamental physical quantities must be understood:

1.1 Voltage (Electric Potential Difference, V)

Voltage, measured in Volts (V), is the electrical pressure or force that pushes electric charges through a conductor. It represents the potential energy difference per unit charge between two points in a circuit. Without voltage, charges remain stationary and no current flows.

1.2 Current (Electric Current, I)

Current, measured in Amperes (A) or Amps, is the rate of flow of electric charge across a cross-section of a conductor over time. One ampere corresponds to one Coulomb of charge passing a point in one second ($I = Q / t$). Current flows from higher electric potential to lower potential in conventional flow.

1.3 Resistance (Electrical Resistance, R)

Resistance, measured in Ohms (Omega), is the opposition that a material presents to the flow of electric current. Factors affecting resistance include:

- Material resistivity (conductors like copper have low resistance, insulators have high resistance)
- Conductor length (longer conductors have higher resistance)
- Cross-sectional area (thicker wires have lower resistance)
- Temperature (resistance typically increases with temperature in metallic conductors)

2. OHM'S LAW AND CIRCUIT MATHEMATICS

2.1 The Statement of Ohm's Law

Formulated by German physicist Georg Simon Ohm in 1827, Ohm's Law states that the current (I) flowing through a conductor between two points is directly proportional to the voltage (V) across the two points and inversely proportional to the resistance (R) of the conductor:

Mathematical Formula:

$$V = I * R$$

$$I = V / R$$

$$R = V / I$$

Where:

- V = Voltage in Volts (V)
- I = Current in Amperes (A)
- R = Resistance in Ohms (Omega)

2.2 Physical Implications & Inversion Rule

- Proportionality: If Voltage doubles while Resistance remains constant, Current will double.
- Inverse Proportionality: If Resistance increases while Voltage remains constant, Current DECREASES proportionally.
Example: If a 12V battery is connected to a 4 Ohm resistor, the current is $I = 12 / 4 = 3$ Amperes. If resistance is increased to 12 Ohms, the current decreases to $I = 12 / 12 = 1$ Ampere.

2.3 The Intuitive Water Pipe Analogy

- Voltage is equivalent to water pressure from a pump or elevated water tank.
- Current is equivalent to the volume flow rate of water through the pipe (gallons per minute).
- Resistance is equivalent to a narrow restriction or valve in the pipe. Narrowing the valve (higher resistance) reduces water flow rate (current).

3. ELECTRICAL POWER AND REVIEW QUESTIONS

3.1 Electrical Power (P)

Power is the rate at which electrical energy is transferred by an electric circuit per unit time, measured in Watts (W).

Formulas:

- $P = V * I$
- $P = I^2 * R$
- $P = V^2 / R$

3.2 Key Conceptual Summary & Takeaways

1. Voltage pushes charges; Resistance opposes charge flow; Current is the resultant rate of flow.
2. Ohm's Law relates them via $I = V / R$.
3. Increasing resistance always reduces current when voltage is held constant.

3.3 Check Questions

Q1: What happens to current if resistance increases while voltage remains constant?

Answer: Current decreases (inverse relationship).

Q2: A 9V circuit has a resistance of 3 Ohms. What is the current?

Answer: 3 Amperes ($I = 9V / 3 \text{ Ohms} = 3A$).